Table 1: Leave-one-out ablation of the two core contributions on CORE4D-G1 (50 sequences; the two E167A box021 cases are dropped and box023 uses the full-stack baseline so all baselines are homogeneous). Each ablation removes one sub-component and re-runs the sampling-based MPC; **Ours** is the full method. *Contribution 1 (Multi-Scale Contact)*: −SB removes the fine SDF surface-band term, −HDMI removes the coarse anchor-attraction term. *Contribution 2 (Part-wise Penetration)*: −gate removes all hard candidate gates, −pen removes all soft penetration penalties. ↑/↓: higher/lower is better; * marks a paired Wilcoxon difference from Ours at $p < 0.05$. Note −HDMI's *lower* penetration is an artifact of failing to reach the object (see EEF pos err), not better contact.

Metric	Ours	−SB	−HDMI	−gate	−pen
Phys. contact@3mm ↑	**0.588**	0.263*	0.394*	0.568	0.581
Phys. pen@3mm ↓	**0.174**	0.217*	0.095*	0.187	0.182
Geom. pen@2mm ↓	**0.050**	0.175*	0.039	0.050	0.055
Root pos err (cm) ↓	**13.66**	11.43*	39.54*	18.38*	15.00
Root ori err (deg) ↓	**8.88**	4.94*	5.78*	15.73*	10.93*
EEF pos err (cm) ↓	**13.07**	11.85*	40.01*	16.49*	14.32*
EEF ori err (deg) ↓	**17.21**	13.05*	9.65*	24.77*	19.12*
Obj pos err (cm) ↓	**10.34**	9.74*	10.24	10.74	10.48
Obj ori err (deg) ↓	**5.11**	4.99	4.97	5.47*	5.60*
Obj speed max (m/s) ↓	**1.29**	1.35*	1.26	1.31	1.32
Foot slip max (m) ↓	**0.87**	0.91	0.83	0.86	0.81